



THE COPPERBELT UNIVERSITY
SCHOOL OF INFORMATION AND COMMUNICATION
TECHNOLOGY

COMPUTER SCIENCE DEPARTMENT

CS301 THIRD YEAR COMPUTER SCIENCE GROUP PROJECT

TITLE: BOOKMYBUS ZAMBIA MANAGEMENT SYSTEM

GROUP: 29

NAME	SIN	ROLES	SIGNATURE
MULENDA FAVOUR	23124723	CHIEF PROGRAMMER	
ZULU MAPALO	22110075	QUALITY ASSURANCE MANAGER	
SIYONDA PELEKELO MAKEKE	23137589	DOCUMENTATION MANAGER	
KATANGA JOHN	23115783	PROJECT MANAGER	

SUPERVISOR APPROVAL:

NAME: MR. MUKOKWEZA

SIGNATURE: _____

DATE: _____

1 Contents

1. ABSTRACT	1
2. INTRODUCTION	2
3. BACKGROUND	3
4. PROBLEM STATEMENT	4
5. OBJECTIVES	5
5.1. SPECIFIC OBJECTIVES	5
6. HYPOTHESIS	6
7. BENEFITS/AIMS OF THE PROJECT	7
8. PURPOSE, SCOPE & APPLICABILITY	8
8.1. PURPOSE	8
8.2. SCOPE	8
8.3. APPLICABILITY	8
9. LITERATURE REVIEW	9
9.1 INTRODUCTION	9
9.2 RELATED WORKS	9
9.3 EXISTING SYSTEMS REVIEW	11
9.3.1 GIG Mobility (Nigeria)	11
9.3.2 Digital Matatu Project (Kenya)	11
9.3.3 KaCyber (Uganda)	12
9.3.4 ViserBus / African Bus (Zambia)	12
9.3.5 Bus Ticket Booking Africa Platform	13
9.3.6 RedBus (India)	13
9.3.7 EasyCoach Online Booking (Kenya)	14
9.3.8 Intercap Bus Online (South Africa)	15
9.3.9 Greyhound e-Ticketing (Southern Africa)	15
9.3.10 Shimomba et al. (2025) — Integrated Online Bus Ticketing (Zambia)	16
9.4 CONCLUSION	16
10. RESEARCH METHODOLOGY	17
10.1. INTRODUCTION	17
10.2. SDLC METHODOLOGY: AGILE SCRUM	17
10.3. OOAD APPROACH WITH UML	17
10.4. DATA COLLECTION	17
10.5. SYSTEM DESIGN	17
10.6. DEVELOPMENT & TESTING	18

10.7.	PHASED IMPLEMENTATION.....	18
10.8.	SECURITY & VALIDATION	18
10.9.	DEPLOYMENT & EVALUATION.....	18
11.	SIGNIFICANCE	19
12.	EXPECTED CONTRIBUTIONS AND IMPLICATIONS.....	20
13.	ETHICAL CONSIDERATIONS.....	21
14.	PROJECT TIMELINE.....	22
15.	FINANCIAL IMPLICATIONS	23
16.	CONCLUSION.....	24
2	REFERENCES.....	25

1. ABSTRACT

In Zambia, Inter-province bus travel is inefficient due to the absence of a centralized digital system. Travelers must physically visit stations to compare fares and book tickets, while operators rely on manual processes that limit reach and convenience.

The proposed BOOKMYBUS ZAMBIA MANAGEMENT SYSTEM will provide a web-based platform where operators can publish fares and seat availability, and travellers can search routes, compare prices, select seats, and make secure online payments. This solution enhances transparency, convenience, and efficiency, modernizing Zambia's bus transport sector for both passengers and operators.

2. INTRODUCTION

Transportation systems play a vital role in supporting economic activities and social mobility as it enables movement of people across regions. In Zambia, road-based public transport remains the primary means of long-distance travel due to its affordability and accessibility compared to other modes such as airplane and train. As populations grow and travel demands increase, the need and importance for efficient, reliable and transparent transport services increase as well.

Advances in Information and Communication Technologies (ICTs) have transformed service delivery in sectors such as banking, education and commerce by improving access to information and automating certain transactions. In the transport sector, digital platforms have been used globally to provide fare information, enable online ticket booking and enhance convenience for the customer. However, the adoption of such technologies within Zambia's inter-province bus transport sector remains restricted.

This study focuses on the development of a BookMyBus Zambia Management System aimed at improving fare transparency and ticket accessibility.

3. BACKGROUND

In Zambia, Inter-province bus transport is one of the most affordable and widely used means of long-distance travel. It serves potential travellers such students, workers, traders, tourists and others. Despite its importance, the sector continues to rely heavily on manual processes for inquiring fare prices and ticket sales.

Travelers are often required to physically visit bus stations when trying to obtain fare price information and purchase of tickets, while bus operators manage seat allocation and sales using paper-based methods. As demand for convenient, reliable transport services grows, traditional approaches have become increasingly inadequate in meeting user expectations.

With the expansion of internet access, smartphone usage and mobile money services in Zambia, there's a growing potential for the application of Information and Communication Technology (ICT) to improve service delivery in the transport sector. This situation highlights the need for a centralized digital platform to support fare price access and ticket booking.

4. PROBLEM STATEMENT

Zambia's inter-province bus transport sector continues to rely heavily on manual fare price inquiry, ticket sales and seat allocation. There is no centralized digital platform through which travellers can access accurate fare information or book tickets remotely. Consequently, passengers are often required to physically go to a bus station to compare prices, check bus availability and purchase tickets, often spending unnecessary time and money.

Research from other countries shows that when bus companies don't use digital systems, it creates several problems. For passengers, prices can be unpredictable and hard to compare, making travel less fair and accessible. This is especially difficult for people who find it hard to travel to a station, are on a tight schedule or budget, or live far from city centres. For the bus companies themselves, doing everything by hand is inefficient. It makes managing seat availability, preventing double bookings, and keeping good records very challenging. It also limits their ability to reach customers who cannot physically come to the bus station, holding back their business growth.

Despite growing availability of the internet and widespread use of mobile money services in Zambia, these technologies haven't been adequately taken advantage of to modernize inter-province bus ticketing. This study seeks to investigate how the lack of an integrated digital fare price tracking and booking platform affects fare price transparency, service accessibility and operational efficiency in Zambia's inter-province bus transport sector.

5. OBJECTIVES

To develop a centralized bus fare tracking and online booking system that improves fare transparency and ticketing accessibility for travellers and operators in Zambia.

5.1. SPECIFIC OBJECTIVES

- Design and implement a web-based bus fare and booking system.
- Implement and analyse the functionality of an operator management portal enabling registered bus companies to publish, update, and manage dynamic fare structures and schedules, administer real-time seat inventory and availability, monitor, review, and analyse passenger booking trends and data. To allow travellers to search routes, compare fare prices, select seats, and securely book and pay for tickets online.
- Onboard a minimum of 5 bus operators and enable booking across 20 major routes during the pilot phase.
- Replace the need for manual physical station visits by providing a complete online platform for fare inquiry and ticket purchase.
- Engineer an automated notification system used to deliver the digital tickets to users via email.

6. HYPOTHESIS

By creating BookMyBus Zambia Management System, a website where people can check bus prices and book tickets online, bus travel in Zambia will be much easier, fairer, and more modern for everyone.

The following outcomes are anticipated:

Prices will become clearer and fairer: Right now, people have to go to the bus station to ask for prices, which can vary from one operator to another. With our system, all prices will be shown online, in one place. This will help passengers choose the best deal and encourage bus companies to offer fair prices.

Booking tickets will become quick and simple: Instead of standing in long queues at the station, travellers will be able to book and pay for their seat from their phone or computer anytime, anywhere. This will save people time and reduce stress.

Bus operators will reach more customers and earn more: Small and large bus companies alike will be able to advertise their trips online and sell tickets to people who can't make it to the station. This means more passengers, fewer empty seats, and better cash flow for operators.

More people will use and trust the system: Because we're designing the platform to work even with slow internet and allowing payment through mobile money (like Airtel Money and MTN Money), we expect both city and rural users to adopt it easily. We also believe they'll be happier with their booking experience compared to the old way.

The whole transport system will move into the digital age: The platform won't just help passengers and bus companies, it will also provide useful data to government and transport authorities. This can help plan better routes, improve services, and support Zambia's goal of becoming a more digitally connected country.

7. BENEFITS/AIMS OF THE PROJECT

This project aims to modernize Zambia's inter-province bus sector through digitalization, delivering key benefits to all stakeholders.

For Travelers (Commuters): Provides convenience through 24/7 online fare comparison and booking from anywhere, leading to financial savings from informed price choices. Digital tickets and seat selection enhance the travel experience.

For Bus Operators: Delivers operational efficiency by automating fare updates and booking management, expanding their customer reach beyond physical stations, and improving cash flow with guaranteed pre-payments.

For the Industry & Authorities: Generates strategic insights from aggregated travel data to aid in planning and policy, while promoting transparency and standardization across the transport sector.

8. PURPOSE, SCOPE & APPLICABILITY

8.1. PURPOSE

To address critical inefficiencies in Zambia's inter-province bus transport by developing a centralized digital platform. This system will solve the lack of accessible fare information and manual ticketing processes that currently force travellers to make physical station visits and limit operator reach.

8.2. SCOPE

The project covers scheduled inter-province and major inter-town bus services in Zambia.

Core functionalities include:

- Operator Portal: Registration, fare/schedule publishing, seat inventory management, booking oversight
- Traveler Interface: Route search, fare comparison, seat selection, secure online payment, digital ticket receipt
- Admin Backend: Operator verification, system monitoring, data integrity management

Specific Exclusions

- Real-time GPS bus tracking
- Multi-modal transport integration
- In-trip service management
- Advanced analytics suites
- Implementation of SMS/USSD in the system

8.3. APPLICABILITY

- Travelers: Easier fare comparison and remote booking
- Bus Operators: Digitized sales, reduced admin work, wider reach
- Transport Authorities: Access to aggregated data for oversight and sector insights
- This platform modernizes the booking experience while providing foundational data for transport sector development.

9. LITERATURE REVIEW

9.1 INTRODUCTION

The development of digital transport systems has gained significant attention globally, particularly in the context of fare price transparency and online ticket booking in developing countries. This literature review incorporates findings from recent studies (2016 – 2025) that inform the design and implementation of a web-based fare tracker and booking system for Kitwe, Zambia.

9.2 RELATED WORKS

Regional analyses of public transport systems in Southern Africa indicate that fragmented ticketing processes and limited digital integration continue to undermine passenger convenience and service efficiency (African News Agency, 2025).

Digital fare payment and e-ticketing adoption are critical to enhancing passenger convenience and operational efficiency. Oluwaseun and Adebayo (2022), in their study of GIG mobility (an online bus ticket booking platform) in Nigeria, highlighted that simplified interfaces significantly increase operator and passenger adoption, especially among users with low digital literacy. Similarly, the Digital Matatu Project in Nairobi (Williams et al., 2015) demonstrated that mobile technologies can improve route visibility and fare consistency even in semi-formal transport sectors.

This suggests that a centralized web-based platform in Zambia could significantly reduce manual fare price inquiries while improving transparency and accessibility for students and commuters.

Mobile payment adoption plays a pivotal role in digital system success. Research such as “Transit Made Easy: Mobile Fare Payment Adoption” (2024) and Mwangi & Chepkwony (2023) found that integrating mobile money services dramatically increased user adoption, whereas card only systems suffered low uptake. Recent evidence from developing cities further demonstrates that digital payment adoption in public transportation is influenced by travel behaviour patterns and mode choice, outlining the importance of flexible and inclusive payment options in system design (Wani et al., 2025).

This suggests that in Zambia, incorporating mobile money payment methods can increase usability and adoption among students.

Technical architecture and system implementation have been widely studied. Kuma et al. (2020) proposed offline-first architectures to ensure high transaction completion in areas with low-connectivity, while Patel & Singh (2023) emphasized temporal database designs for managing dynamic fare adjustments. Agyeman et al. (2020) documented hybrid SMS/USSD and web-based solutions that reduce data requirements while delivering real-time information. Security considerations are critical in online transport booking platforms. Studies on transport payment systems in developing economies emphasize that compliance with standards such as PCI DSS is essential for protecting user data and building trust among passengers and operators (Banda et al., 2021).

These studies provide guidance for designing a web-based system that functions effectively under variable internet conditions while supporting real-time fare prices and seat availability.

User experience is another critical factor. Ngoma et al. (2019) demonstrated that icon-based navigation, voice prompts and multilingual support significantly enhanced task completion rates among novice users in Zambian areas around urban settings. Schmidt & Kalinda (2022) highlight the importance of balancing needs of multiple stakeholders, including passengers, operators and regulators. Designing interfaces that accommodate students' level of digital literacy and provide intuitive booking workflows will be essential for adoption in Kitwe.

African case studies provide context-specific insights. Mensah (2021) showed that digital fare information in Ghana reduced price variation and overall passenger costs. The World Bank (2022) reported that mobile money integration is key to successful adoption in African urban transport systems.

Locally, studies such as “Design and Development of Integrated Online Bus Ticketing” (2025) and “Study of Public Transport in Lusaka” (2022) demonstrates how feasible web-based ticketing is, while calling attention to the limited adoption of such platforms due to low awareness and connectivity challenges. In addition, research on the digitisation of Zambian intercity bus services shows that while digital ticketing solutions are technically feasible, adoption has remained limited due to low awareness and infrastructure limitations (Study on digitisation of Zambian intercity bus services, 2025).

Despite this growing body of research, several gaps remain that the proposed system aims to address. There are limited studies on inter-province transport digitalization in Zambia, especially integrating fare transparency with online booking capabilities.

9.3 EXISTING SYSTEMS REVIEW

9.3.1 GIG Mobility (Nigeria)

GIG Mobility is one of Nigeria's leading digital bus booking platforms. It offers online ticket reservations, mobile app access, seat selection, and real-time schedules. Its similarity to BookMyBus Zambia lies in its focus on convenience and digital adoption in a developing country context. The key lesson is that simplified interfaces and mobile-first design significantly increase adoption among users with low digital literacy (GIG Mobility,2023).

Features:

- Online ticket reservations
- Mobile app access
- Seat selection
- Real-time schedules

Advantages

- Simplified, user-friendly interface
- Mobile-first design increases accessibility
- Focus on convenience and digital adoption

Disadvantages

- Limited offline functionality
- May struggle in areas with poor internet connectivity

9.3.2 Digital Matatu Project (Kenya)

The Digital Matatu Project digitized Nairobi's informal transport sector by mapping routes, fares, and schedules into mobile-accessible formats. While not a booking system per se, it addresses fare transparency and route visibility. The similarity is its effort to bring order and fairness to fragmented transport systems. The lesson is that transparency in fares and routes builds trust and accessibility, especially in semi-formal transport markets (Williams et al, 2015).

Features

- Digitized route, fare, and schedule mapping

- Mobile-accessible formats

Advantages

- Increases fare and route transparency
- Builds trust in informal transport systems
- Enhances accessibility and fairness

Disadvantages:

- · Not a booking system—limited transaction capability
- -Relies on continuous data updates

9.3.3 KaCyber (Uganda)

KaCyber provides online ticketing for buses, trains, and ferries, with strong integration of mobile money services. Its similarity to BookMyBus Zambia is the reliance on mobile money as the primary payment method, which is critical in regions where card adoption is low. The lesson is that flexible payment options, especially mobile money, are essential for widespread adoption (KaCyber,2022).

Features:

- · Online ticketing for multiple transport modes
- · Mobile money integration

Advantages:

- · Supports low-card-adoption regions
- · Flexible payment options increase accessibility
- · Multi-modal transport integration

Disadvantages:

- · Dependent on mobile money network stability
- · Limited payment method variety

9.3.4 ViserBus / African Bus (Zambia)

ViserBus is a localized Zambian platform offering online ticket purchases, seat preferences, and e-ticket delivery. It directly parallels BookMyBus Zambia by addressing the same market and challenges. The lesson here is that localized branding and trust-building are crucial for adoption, as users are more likely to trust platforms that feel home grown (ViserBus, 2024).

Features:

- · Online ticket purchases

- · Seat preference selection
- · E-ticket delivery

Advantages:

- · Localized branding builds trust
- · Directly addresses Zambian market needs
- · User-friendly booking process

Disadvantages:

- · Limited regional or cross-border coverage
- · Smaller operator network compared to larger platforms

9.3.5 Bus Ticket Booking Africa Platform

This regional platform covers multiple African countries, offering real-time GPS tracking, Wi-Fi, and digital ticket downloads. Its similarity lies in providing centralized access to multiple operators. The lesson is that value-added services such as GPS tracking and onboard Wi-Fi can differentiate a system and enhance customer experience beyond basic ticketing (Bus Ticketing Book Africa,2023).

Features:

- · Multi-country coverage
- · Real-time GPS tracking
- · On-board Wi-Fi access
- · Digital ticket downloads

Advantages:

- · Centralized access to multiple operators
- · Value-added services improve customer experience
- · Regional scalability

Disadvantages:

- Requires reliable GPS and internet infrastructure
- May be complex to manage across different regions

9.3.6 RedBus (India)

RedBus is Asia's largest bus booking platform, aggregating thousands of operators into one system. It offers route search, seat selection, and mobile payments. The similarity is its centralized approach to fare transparency and booking. The lesson is that aggregation of

multiple operator's builds scale, improves user trust, and creates a one-stop solution for travellers (RedBus, 2012).

Features:

- · Aggregation of multiple bus operators
- · Route search and seat selection
- · Mobile payment integration

Advantages:

- · Large scale improves trust and reliability
- · One-stop solution for travellers
- · Transparent fare and booking system

Disadvantages:

- · Can be overwhelming for users due to volume of options
- · Dependent on operator cooperation and data accuracy

9.3.7 EasyCoach Online Booking (Kenya)

EasyCoach is a major Kenyan bus operator that digitized its ticketing system with web and mobile booking, fare transparency, and SMS ticket confirmations. Its similarity to BookMyBus Zambia is the intercity bus context with digital ticketing. The lesson is that SMS confirmations are particularly useful in areas with limited internet access, ensuring inclusivity (EasyCoach, 2023).

Features:

- · Web and mobile booking
- · Fare transparency
- · SMS ticket confirmations

Advantages:

- · SMS confirmations aid inclusivity in low-internet areas
- · Straightforward digital ticketing for intercity travel
- · Builds operator-customer trust

Disadvantages:

- · Limited to one operator's services
- · SMS dependency may incur extra costs

9.3.8 Intercape Bus Online (South Africa)

Intercape is one of Southern Africa's largest bus operators, offering online booking, seat maps, and cross-border routes. Its similarity lies in digitizing ticketing across multiple regions. The lesson is that integration with cross-border travel expands reach and positions the platform as a regional leader rather than just a local solution (Intercape, 2024).

Features:

- · Online booking
- · Seat maps
- · Cross-border route integration

Advantages:

- · Expands market reach through cross-border services
- · Enhances user convenience with seat mapping
- · Positions platform as regional leader

Disadvantages:

- · Regulatory complexities across borders
- · Higher operational and logistical challenges

9.3.9 Greyhound e-Ticketing (Southern Africa)

Greyhound operates across South Africa, Zimbabwe, and Zambia, offering online booking, mobile payments, and printable tickets. Its similarity is that it already functions in Zambia, showing demand for digital ticketing. The lesson is that established operators can drive adoption when they digitize, making partnerships with such companies a strategic move (Greyhound, 2023).

Features:

- · Online booking
- · Mobile payments
- · Printable e-tickets

Advantages:

- · Demonstrates existing demand in Zambia
- · Established operator trust aids adoption
- · Multi-country presence

Disadvantages:

- · Limited to Greyhound services only
- · May not integrate with other operators

9.3.10 Shimomba et al. (2025) — Integrated Online Bus Ticketing (Zambia)

This academic project proposed a web-based ticketing system with mobile money integration, similar to BookMyBus Zambia. Its similarity is direct, as it addresses the same Zambian context and challenges. The lesson is that while feasibility is proven, adoption barriers such as low awareness and connectivity must be addressed through user education and offline-first design (Shimomba et al, 2025).

Features:

- · Web-based ticketing
- · Mobile money integration

Advantages:

- · Directly addresses Zambian transport challenges
- · Proves feasibility of digital ticketing in local context
- · Mobile money integration aligns with local payment habits

Disadvantages:

- · Still in proposal/academic stage—not commercially deployed
- · Adoption barriers like low digital literacy and connectivity remain

9.4 CONCLUSION

From these studies, several design implications for the proposed system emerge: The platform must be user-friendly and accessible, incorporate mobile money payment options, allow for offline first capabilities to mitigate connectivity issues, include operator dashboards to manage fares, seat availability and bookings. In addition, it must be scalable and reliable, capable of supporting multiple routes, real-time updates and a growing student and operator user base.

10. RESEARCH METHODOLOGY

10.1. INTRODUCTION

This chapter outlines the methodology for developing BookMyBus Zambia Management System using Agile Scrum with Object-Oriented Analysis and Development (OOAD).

10.2. SDLC METHODOLOGY: AGILE SCRUM

The project follows Agile Scrum with two-week sprints to ensure iterative development and continuous stakeholder feedback. This allows incremental delivery—starting with fare tracking, then adding booking functionality—while adapting to Zambia's evolving transport needs.

10.3. OOAD APPROACH WITH UML

System design uses OOAD principles modelled with UML:

- Use Case Diagrams for Travelers, Operators, Administrators
- Class Diagrams for core objects (Route, Booking, Payment)
- Sequence Diagrams for booking and payment flows
- State Diagrams for ticket lifecycle

10.4. DATA COLLECTION

- Primary: Structured interviews with Zambian bus operators; surveys at Kitwe Town Bus Terminal
- Secondary: Analysis of existing fare sheets; technical review of Zambian payment gateways (Zanaco Kwacha, Airtel Money)

10.5. SYSTEM DESIGN

- Tier Architecture: HTML, CSS, and JavaScript (React.js) frontend; PHP (Laravel framework) backend; PostgreSQL database; Firebase for hosting and authentication
- Backend Rationale: PHP/Laravel provides robust server-side logic, security features, and RESTful API development. PostgreSQL offers relational data integrity for

transactional booking systems. Firebase enables scalable hosting and real-time authentication.

- Design Tools: Figma wireframes, system architecture diagrams, ER diagrams

10.6. DEVELOPMENT & TESTING

Technology Stack:

- Frontend: HTML/CSS/React.js
- Backend: PHP (Laravel framework)
- Database: PostgreSQL
- Services: Firebase (hosting, authentication), Zambian payment gateway APIs

10.7. PHASED IMPLEMENTATION

Phase 1: Operator registration & fare management

Phase 2: Booking engine (seat selection, user accounts)

Phase 3: Payment integration & digital ticket generation

Phase 4: Admin dashboard & system optimization

Testing: Unit testing (Jest/React Testing Library, PHPUnit), integration testing, system testing, UAT with Zambian operators, security and performance testing.

10.8. SECURITY & VALIDATION

- Firebase Authentication for role-based access control
- PCI-DSS compliant payment gateway integration
- PostgreSQL transactions for data integrity
- Real-time seat inventory validation
- Comprehensive audit logging

10.9. DEPLOYMENT & EVALUATION

- Hosting: Firebase hosting with CDN optimization
- Pilot: 5 operators on 20 major Zambian routes
- Metrics: Uptime (>90%), booking success rate (>95%), page load time (<5s)
- Impact Assessment: Reduction in physical station inquiries, operator efficiency gains

11. SIGNIFICANCE

The project fills a major gap in Zambia's transport sector by creating the first integrated digital platform for fare transparency and online bus booking. It benefits travellers with convenient, cost-effective trip planning, supports operators through digitized sales and wider reach, and promotes digital inclusion across urban and rural areas—aligning with Zambia's digital transformation agenda.

- The users no longer have to visit the terminal to see if a bus is available
- Providing a visual layout of the bus for users to choose their preferences
- Checking whether a preferred seat is available
- Allow 24/7 booking services

12. EXPECTED CONTRIBUTIONS AND IMPLICATIONS

This project offers transformative value to Zambia by addressing a critical gap in the nation's transportation infrastructure. It introduces the first integrated digital platform specifically designed for Zambia's inter-province bus sector, directly tackling the daily inconvenience faced by travellers who currently rely on physical station visits for fare information and ticket purchases.

For **Zambian Travelers**, the system delivers tangible time and cost savings. It eliminates the need for expensive, time-consuming trips to bus terminals just to find out the prices. Citizens across all provinces gain the ability to compare real-time fares, plan journeys efficiently, and book tickets securely using locally trusted payment methods such as Zanaco Kwacha, Airtel Money, and MTN Mobile Money. This digital access is particularly transformative for rural communities, budget-conscious families, and individuals with mobility constraints, democratizing access to travel planning.

For **Zambian Bus Operators**, the platform provides a practical pathway to digital transformation. Companies can modernize their manual, paper-based ticketing systems, reduce cash-handling risks, and minimize administrative overhead. The system expands their market reach beyond physical terminals, potentially increasing passenger volumes while providing valuable insights into booking patterns and customer demand—enabling more responsive and competitive service offerings.

For **Zambia's Broader Development**, this initiative demonstrates how locally contextualized technology can drive sectoral modernization. It aligns with national priorities around digital inclusion, financial technology adoption, and transport efficiency. The aggregated, anonymized travel data generated by the platform could inform evidence-based policy making, infrastructure planning, and regulatory oversight, contributing to a more transparent, efficient, and user-centred national transport network.

13. ETHICAL CONSIDERATIONS

Key ethical issues include data privacy (sensitive user/payment information), informed consent, algorithmic fairness in fare displays, and digital exclusion risks. These will be mitigated through: adherence to the Zambia Data Protection Act; use of PCI-DSS compliant payment gateways (e.g., Zanaco Kwacha); transparent terms and consent processes; unbiased, clear listing criteria for operators; and accessible design for users with limited connectivity or digital literacy. The project will follow the ACM Code of Ethics, anonymize research data, and ensure no hidden fees or biased prioritization.

14. PROJECT TIMELINE

TASK \ MONTH	JANUARY				FEBRUARY				MARCH				APRIL				MAY				JUNE				JULY				AUGUST			
	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4
PROPOSAL DRAFTING																																
LITERATURE REVIEW																																
SYSTEM DESIGN																																
DEVELOPMENT & TESTING																																
SYSTEM IMPLEMENTATION																																
SYSTEM DOCUMENTATION																																

15. FINANCIAL IMPLICATIONS

Item	Estimated Cost (ZMW)
Domain & Hosting	300
Testing Devices & Data Bundles	400
Research & Data Collection	500
Printing & Branding	500
Total	1 700

16. CONCLUSION

This project has developed and proposed a practical digital solution to modernize Zambia's inter-province bus transport sector.

The proposed "BookMyBus Zambia Management System" directly addresses the core inefficiencies of manual fare inquiry and ticket purchasing by introducing a centralized web-based platform. It is specifically designed for the local context, prioritizing user-friendly access, integration with ubiquitous mobile money services, and robust functionality for both travellers and operators.

By enabling real-time fare comparison, remote seat selection, and secure online payment, the system will eliminate the need for time-consuming and costly physical trips to bus terminals. For operators, it provides a vital tool to digitize sales, manage inventory efficiently, and expand their customer base beyond station premises.

The platform thus acts as a critical bridge, connecting supply with demand through transparent and convenient digital channels. The successful implementation of this system will deliver significant benefits: enhancing convenience and cost-saving for passengers, improving operational reach and efficiency for bus companies, and generating valuable data for sector planning.

Ultimately, this project concludes that the adoption of this integrated platform represents a necessary and achievable step forward. It will transform the user experience, drive sectoral digitization, and contribute meaningfully to a more efficient, transparent, and modern public transport system in Zambia.

REFERENCES

- African News Agency. (2025). Digital solutions for South Africa's passenger transport crisis: Lessons from global innovation. <https://ai-impact.co.za/digital-solutions-for-south-africas-passenger-transport-crisis-lessons-from-global-innovation>
- Agyeman, O. G., Oloke, O. C., & Oyedepo, S. O. (2020). Evolution of a secure system architecture for bus ticket booking: Dual platform approach. *Journal of Science, Technology, Mathematics and Education (JOSTMED)*, 16(2), 114–124. <https://jostmed.futminna.edu.ng/index.php/jostmed/article/view/412>
- American Public Transportation Association. (2024). Transit made easy: Examining the adoption and impact of mobile fare payment technology among bus riders. <https://www.sciencedirect.com/science/article/pii/S2590198224000721>
- Banda, L., Phiri, J., & Zulu, M. (2021). PCI DSS compliance for small transport operators in developing economies: A Zambian case study. In *Proceedings of the 2021 IEEE African Conference (AFRICON)* (pp. 1–6). IEEE.
- Bus Ticketing Book Africa. (2023). About our regional transport platform: GPS tracking and digital ticket solutions. <https://busticketbookafrica.com/about>
- EasyCoach Ltd. (2023). Online booking and digital ticketing services. <https://easycoach.co.ke/>
- GIG Mobility (GIGM). (2023). Tech-driven mobility for intercity travel in Nigeria. <https://gigm.com>
- Greyhound Coaches South Africa. (2023). Digital transformation and passenger services in Southern Africa. <https://greyhound.co.za/>
- Intercape South Africa. (2024). Intercape Mainliner and Sleepliner: Regional bus travel. <https://www.intercape.co.za/>

- Kuma, S., & Anbanandam, R. (2020). Blockchain technology in the supply chain: A review of connectivity and digital technology. *Journal of Global Operations and Strategic Sourcing*, 13(3), 223-245. <https://doi.org/10.1108/JGOSS-06-2019-0044>
- Kacyber Uganda (2025). Online Bus Ticket Booking. <https://kacyber.com>
- Mensah, J. K. (2021). Price dispersion in informal transport markets: Evidence from Ghanaian tro-tros. *Transportation Research Part A: Policy and Practice*, 152, 180–195.
- Mwangi, J., & Chepkwony, J. (2023). Barriers to digital payment adoption in East African transport. *Journal of FinTech in Emerging Economies*.
- Ngoma, M., Mulwanda, M., & Banda, C. (2019). Designing for first-time smartphone users in Zambian urban and peri-urban areas: A UX study. *International Journal of Human-Computer Interaction*, 35(20), 1902–1917.
- Oluwaseun, F., & Adebayo, O. (2022). Enhancing passenger convenience through digital adoption: A study of GIG mobility. *Transport Research Board*. <https://archive.org/details/gig-mobility-study-2022>
- Patel, R. S., & Singh, K. (2023). Temporal database designs for managing dynamic fare adjustments in real-time transit systems. *International Journal of Computer Science and Information Technology*, 15(1), 45-58. <https://doi.org/10.5121/ijcsit.2023.15104>
- Public–Private Infrastructure Advisory Facility. (2022). Study of public transport in Lusaka: Final report. [https://ppiaf.org/sites/default/files/documents/2022-01/final_report_lusaka_public_transport_improvement_study_\(1\).pdf](https://ppiaf.org/sites/default/files/documents/2022-01/final_report_lusaka_public_transport_improvement_study_(1).pdf)
- redBus India. (2012). How aggregation revolutionized the Indian bus industry. <https://www.redbus.in/info/aboutus>
- Schmidt, S. S., & Kalinda, T. (2022). Stakeholder needs and digital literacy in Zambian urban transport. Indaba Agricultural Policy Research Institute (IAPRI). <https://www.iapri.org.zm/publications/technical-reports/>
- Shimomba, H., Mupeta, M., & Moonga, S. (2025). Design and development of an integrated online bus ticketing system. *International Journal of Advanced Multidisciplinary Research and Studies*, 5(1), 1288–1294. <https://doi.org/10.62225/2583049X.2025.5.1.3796>

- Study on digitisation of Zambian intercity bus services. (2025). SAPub Journal of Computer [A Study on Digitisation of Zambian Intercity Bus Based Public Transport Support Services](#)
- ViserBus. Online Bus Ticket Booking. [ViserLab - 360 Degree Solution For Your Digital Business](#)
- Wani, S. A., Pani, A., Mohan, R., & Bhowmik, B. (2025). Digital payment adoption in public transportation: Mediating role of mode choice segments in developing cities. *Transportation Research Part A: Policy and Practice*, 191, 104319. <https://doi.org/10.1016/j.tra.2024.104319>
- Williams, S., Klopp, J. M., White, A., Waiganjo Wagacha, P., & Xu, W. (2015). The digital matatu project: Using cell phones to create an open source data for Nairobi's semi-formal bus system. *Journal of Transport Geography*, 49, 39-51. <https://doi.org/10.1016/j.jtrangeo.2015.10.005>
- World Bank. (2022). Innovation in fare payment systems for African cities. [Innovation in Fare Payment Systems for Public Transport in African Cities](#)